

STUDENT ACADEMIC ADVISOR

Fuzzy Rule-Based Expert System and K-Means Clustering

Final Project Report

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Executive Summary

This project combines two complementary AI approaches within one Student Academic Advisor domain. Part A produces transparent support signals for an individual student using Production Rules, O-A-V fuzzy facts, deterministic forward chaining, and a live numerical trace. Part B discovers population-level profiles using a reproducible K-Means pipeline. The two parts remain independent during computation and are compared only after both have run.

The report is deliberately organized around the official grading distribution. Every non-individual rubric item is followed by concrete repository evidence, output evidence, a mathematical example, or an implementation path. Visual proof is used where the project produces a visual artifact; otherwise exact files, tests, and results are stated.

Part A - Knowledge Representation	2	Production Rules, O-A-V schema, rules table, AST and structure evidence	PASS
Part A - Inference Engine	1	Dependency graph, topological order, cycle handling, forward-chain pseudocode	PASS
Part A - Fuzzy Logic	2	Membership mappings, operators, FV/CF/CV, Maximum/Union, worked trace	PASS
Part A - Inference Network	1	Full official-convention diagram and exact rule mapping	PASS
Part A - Individual	4	Technical evidence complete; viva preparation remains an individual responsibility	PREP
Part B - Data Preparation	2	UCI audit, feature decisions, leakage safeguards, StandardScaler pipeline	PASS
Part B - Model Implementation	2	scikit-learn K-Means, K=2..10 evaluation, deterministic final model	PASS
Part B - Evaluation	1	Elbow/Silhouette chart, K=2 justification, profiles and limitations	PASS
Part B - Individual	5	Notebook and evidence complete; viva explanation remains individual	PREP

Overall technical evidence

PASS | Two distinct knowledge representations: Production Rules and O-A-V facts.

PASS | Six validated rules, nested conditions, chained conclusions, deterministic dependency resolution.

PASS | Fuzzy min/max/1-x, FV/CF/CV propagation, Maximum and Union.

PASS | K-Means preparation, K selection, profiles, outputs, and post-hoc integration.

PASS | 219 automated tests passed on the merged development baseline.

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The official grading order is preserved: Part A, then Part B, followed by integration, reproducibility, limitations, conclusion, references, and evidence inventory. PDF bookmarks provide direct navigation to every major heading.

1. Project Overview and System Structure

The advisor targets early, explainable academic-support signals. It is not a diagnostic system and does not make a final institutional decision. Part A answers: why might this individual student need support? Part B answers: what population-level student profiles emerge without expert labels?

Student Academic Advisor - Tested System Architecture

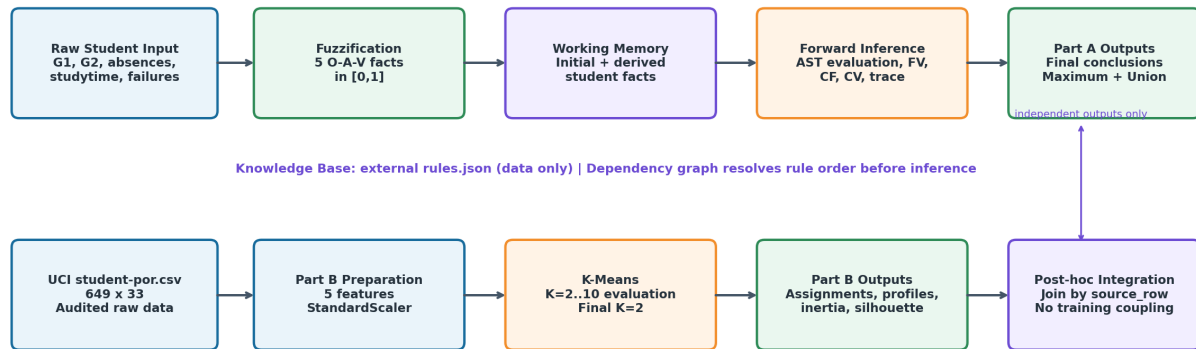


Figure 1. Tested architecture. The Knowledge Base remains data-only; Part A and Part B meet only in post-hoc integration.

1.1 Repository structure

```

student_academic_advisor/
expert_system/ # raw input, fuzzification, models, parser, KB validation,
# dependency graph, fuzzy logic, working memory,
# inference engine, trace, aggregation, demo/report
ml_pipeline/ # preparation, clustering, reporting, reproducible runner
integration/ # independent-output comparison and reporting
notebooks/ # executable Part B presentation notebook
outputs/ # Part A, ML, and integration evidence artifacts
docs/ # decisions, diagrams, audit, results, and final report
tests/ # expert-system, ML-pipeline, and integration tests
  
```

Structural proof

PASS | Knowledge Base: student_academic_advisor/expert_system/knowledge_base/rules.json

PASS | Reusable ML code: student_academic_advisor/ml_pipeline/

PASS | Independent comparison: student_academic_advisor/integration/

PASS | Presentation notebook imports tested modules rather than duplicating the pipeline.

2. Part A - Knowledge Representation

PART A - KNOWLEDGE REPRESENTATION

2 MARKS

VERIFIED

2.1 Technique 1: Production Rules

The first representation technique is an external, human-readable rule set. Each JSON rule has separate id, condition, conclusion, cf, and description fields. The condition excludes IF/THEN and is parsed once into an immutable AST. Domain knowledge is therefore stored as data rather than embedded in the inference engine's control flow.

2.2 Technique 2: O-A-V fuzzy facts

The second technique is Object-Attribute-Value. Each immutable fact is shaped as (student_object, fuzzy_attribute, membership_value), stores a value in [0,1], and records whether it is INITIAL or DERIVED. This is structurally different from Production Rules and maps naturally to one student with many measurable attributes.

O-A-V fact	object_id + attribute + value + origin	POR-0649 + high_absence + 0.25 + INITIAL	State in Working Memory
Production Rule	id + condition + conclusion + cf + description	R1: high_absence OR low_study_time	Static domain knowledge
Condition AST	FACT_REF / NOT / AND / OR nodes	AND(FACT_REF, OR(...))	Unambiguous condition structure

Requirement evidence
PASS Two distinct techniques are implemented: Production Rules and O-A-V fuzzy facts.
PASS Rules are external JSON data; the engine only interprets validated LoadedRule objects.
PASS Fact models enforce real values in [0,1] and explicit INITIAL/DERIVED origin.
PASS Readable descriptions allow a non-programming reviewer to inspect domain knowledge.

2. Part A - Full Production Rule Base

	Condition	Conclusion	Weight	Structural Classification
R1	high_absence OR low_study_time	engagement_concern	0.70	Intermediate
R2	low_first_period_performance AND low_second_period_performance	persistent_low_performance	0.90	Intermediate
R3	persistent_low_performance AND (high_failure_history OR engagement_concern)	core_academic_risk	0.85	Intermediate
R4	high_absence AND NOT (low_first_period_performance OR low_second_period_performance)	attendance_based_support_need	0.60	Final
R5	low_second_period_performance AND NOT low_first_period_performance	emerging_performance_support_need	0.65	Final
R6	core_academic_risk OR (high_failure_history AND low_study_time)	compounded_academic_support_need	0.90	Final

Structural classification is derived, not manually authored. A conclusion is Intermediate when another rule references it; otherwise it is Final. The validated dependency edges are R1 -> R3, R2 -> R3, and R3 -> R6.

2.3 Restricted condition parsing pseudocode

```

PARSE_CONDITION(text):
tokens <- TOKENIZE_RESTRICTED_SYNTAX(text)
ast <- PARSE_OR(tokens)
REQUIRE end_of_input
RETURN ast

PARSE_OR: left <- PARSE_AND; while OR: left <- OR(left, PARSE_AND)
PARSE_AND: left <- PARSE_NOT; while AND: left <- AND(left, PARSE_NOT)
PARSE_NOT: if NOT then RETURN NOT(PARSE_NOT); else RETURN PARSE_PRIMARY
PARSE_PRIMARY: FACT_REF or '(' PARSE_OR ')'
```

Rule-base proof

PASS | Nested/parenthesized rules: R3, R4, and R6.

PASS | NOT precedence is visible in R4 and R5; parser enforces NOT > AND > OR.

PASS | Duplicate IDs, duplicate conclusion producers, unknown facts, collisions, and malformed syntax are rejected.

PASS | No eval() and no hard-coded academic if/else rules are used.

3. Part A - Inference Engine

PART A - INFERENCE ENGINE

1 MARK

VERIFIED

The engine uses deterministic forward chaining. Dependencies are derived from the parsed AST: if Rule A produces an attribute referenced by Rule B, an edge $A \rightarrow B$ is created. Topological sorting runs before inference. Simultaneously available rules are ordered by Knowledge Base declaration index, not alphabetically.

3.1 Dependency resolution pseudocode

```
BUILD_GRAPH(rules):
  producer_by_attribute <- map each conclusion to exactly one rule
  FOR each consumer rule:
    FOR each FACT_REF in consumer.condition_ast:
      IF FACT_REF has a producer:
        add edge producer -> consumer
  RETURN graph

TOPOLOGICAL_ORDER(graph):
  ready <- zero-indegree rules ordered by declaration_index
  WHILE ready is not empty:
    rule <- remove earliest declared ready rule
    append rule to order; reduce successor indegrees
  IF number ordered != number of rules: REPORT exact cycle and STOP
  RETURN order
```

3.2 Forward-chaining pseudocode

```
RUN_INFERENCE(raw_student):
  input <- VALIDATE_RAW_VALUES(raw_student)
  initial_facts <- FUZZIFY(input)
  rules <- LOAD_PARSE_VALIDATE_KB(rules.json)
  order, structural_types <- RESOLVE_DEPENDENCIES(rules)
  memory <- WorkingMemory(initial_facts)
  FOR rule IN order:
    REQUIRE all referenced facts are defined
    fv, operator_steps <- EVALUATE_AST(rule.condition_ast, memory)
    cv <- fv * rule.cf
    derived_fact <- Fact(rule.conclusion, cv, DERIVED)
    memory.assert(derived_fact)
    trace.append(condition, premises, steps, fv, cf, cv, fv>0, cv>0)
  RETURN final_conclusions, Maximum, Union, trace
```

Inference evidence

PASS | Approved order: R1 -> R2 -> R3 -> R4 -> R5 -> R6.

PASS | Intermediate CVs propagate through engagement_concern, persistent_low_performance, and core_academic_risk.

PASS | Self-dependencies and circular paths stop execution before inference.

PASS | Live trace preserves exact evaluation order and numerical operator steps.

4. Part A - Fuzzy Logic

PART A - FUZZY LOGIC

2 MARKS

VERIFIED

Five key facts are converted to concern-oriented fuzzy memberships. Higher membership means greater concern. Raw-to-fuzzy conversion is an approved implementation decision; the official fuzzy requirement is that key facts use values in [0,1].

	Key Fact	Fuzzy Membership
G1	low_first_period_performance	1 if $G1 \leq 9$; $(12-G1)/3$ if $9 < G1 < 12$; otherwise 0
G2	low_second_period_performance	1 if $G2 \leq 9$; $(12-G2)/3$ if $9 < G2 < 12$; otherwise 0
absences	high_absence	0 if ≤ 2 ; $(absences-2)/8$ if $2 < absences < 10$; otherwise 1
studytime	low_study_time	lookup {1:1.0, 2:0.5, 3:0.0, 4:0.0}
failures	high_failure_history	failures / 3 for observed codes 0..3

4.1 Fuzzy operators and conclusion values

```

AND(a,b) = min(a,b)
OR(a,b) = max(a,b)
NOT(a) = 1-a
FV = value of the complete condition AST
CF = author-assigned confidence factor in [0,1]
CV = FV * CF
Maximum = max(Final CVs)
Union(a,b) = a + b - a*b

```

A rule is evaluated when its premises are available, fires exactly when $FV > 0$, and contributes exactly when $CV > 0$. No arbitrary threshold such as 0.5 is introduced. Zero-valued derived facts remain defined because a later NOT may consume them.

4.2 Fuzzification pseudocode

```

FUZZIFY(student):
facts <- empty list
facts += OAV(student.id, low_first_period_performance, LOW_GRADE(student.G1))
facts += OAV(student.id, low_second_period_performance, LOW_GRADE(student.G2))
facts += OAV(student.id, high_absence, HIGH_ABSENCE(student.absences))
facts += OAV(student.id, low_study_time, STUDY_LOOKUP[student.studytime])
facts += OAV(student.id, high_failure_history, student.failures / 3)
RETURN all five facts, including zero-valued facts

```

Fuzzy-logic proof

PASS | Every approved input boundary and membership output is tested.

PASS | FV, CF, and CV retain full internal precision; rounding occurs only for display.

PASS | Both Maximum and pairwise fuzzy Union aggregate structurally Final Conclusions only.

5. Part A - Inference Network Diagram

PART A - INFERENCE NETWORK DIAGRAM

1 MARK

VERIFIED

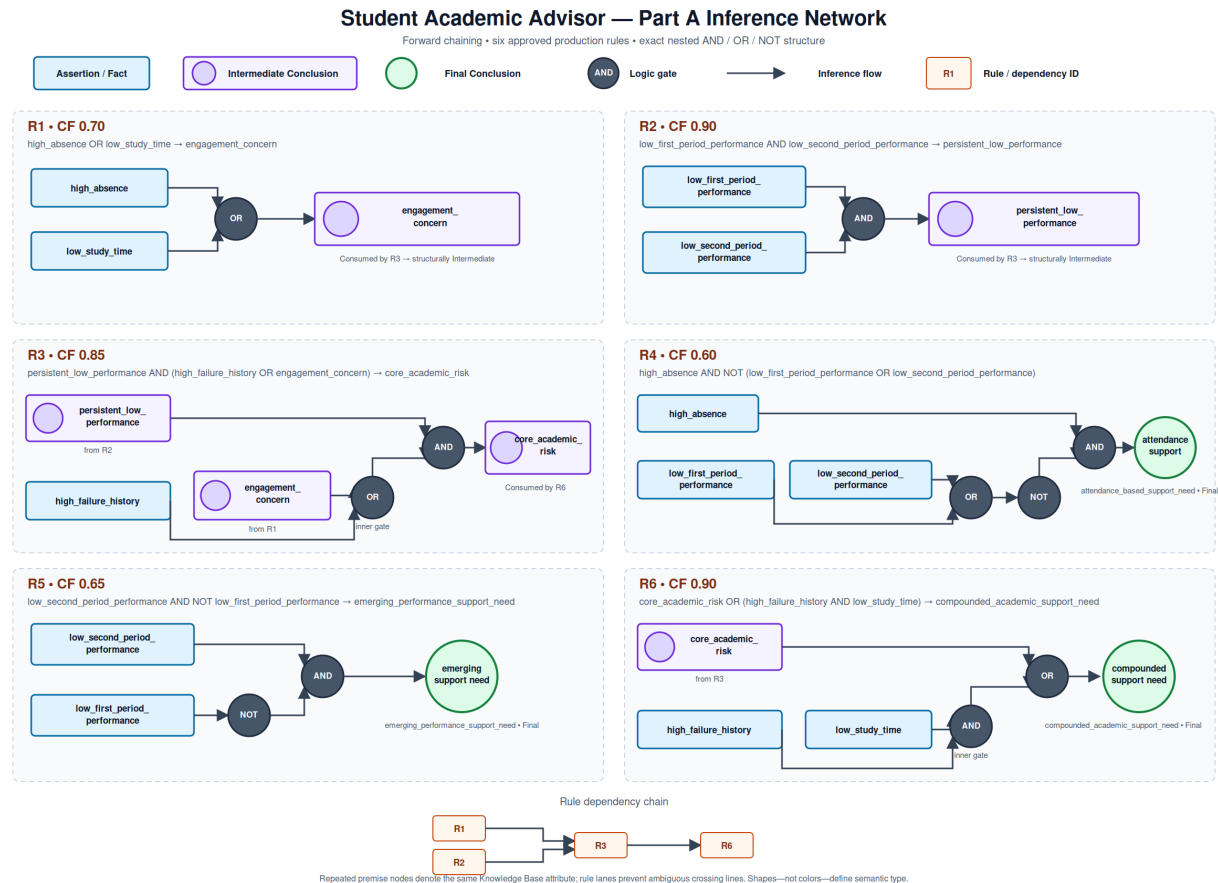


Figure 2. Exact six-rule inference network. Squares are facts; circle-inside-square nodes are Intermediate Conclusions; plain circles are Final Conclusions; labeled gates implement AND/OR/NOT.

5. Part A - Diagram Mapping and Structural Evidence

The diagram is not a decorative redraw. It represents the same six conditions, confidence factors, conclusions, and dependencies used by the Knowledge Base and the live engine.

Rule	Rule expression	Conclusion type	Dependency evidence
R1	high_absence OR low_study_time	Intermediate	consumed by R3
R2	low_first AND low_second	Intermediate	consumed by R3
R3	persistent_low AND (failure OR engagement)	Intermediate	consumed by R6
R4	absence AND NOT (low_first OR low_second)	Final	no consumer
R5	low_second AND NOT low_first	Final	no consumer
R6	core_risk OR (failure AND low_study)	Final	no consumer

Visual convention proof

PASS | Five initial fuzzy facts are drawn as squares.

PASS | Three consumed conclusions are circles inside squares.

PASS | Three unconsumed conclusions are plain circles.

PASS | Nested R3, R4, and R6 groupings and every AND/OR/NOT gate are visible.

PASS | Rule dependency chain at the bottom shows R1/R2 -> R3 -> R6.

6. Part A - Complete Worked Example

The evidence below is generated from the real final record in student-por.csv, identified as POR-0649. It uses the validated input, approved membership functions, loaded Knowledge Base, resolved rule order, and live trace.

6.1 Raw input and initial O-A-V facts

G1	10	low_first_period_performance	$(12-10)/3$	0.666667
G2	11	low_second_period_performance	$(12-11)/3$	0.333333
absences	4	high_absence	$(4-2)/8$	0.250000
studytime	1	low_study_time	lookup category 1	1.000000
failures	0	high_failure_history	0/3	0.000000

6.2 Live trace summary

1	R1	1.000000	0.70	0.700000	Yes	engagement_concern = 0.700000
2	R2	0.333333	0.90	0.300000	Yes	persistent_low_performance = 0.300000
3	R3	0.300000	0.85	0.255000	Yes	core_academic_risk = 0.255000
4	R4	0.250000	0.60	0.150000	Yes	attendance_based_support_need = 0.150000
5	R5	0.333333	0.65	0.216667	Yes	emerging_performance_support_need = 0.216667
6	R6	0.255000	0.90	0.229500	Yes	compounded_academic_support_need = 0.229500

All six rules are evaluated because their premises become available. All fire because $FV > 0$, and all contribute because $CV > 0$. This is a property of this input, not a claim that every student fires every rule.

Worked-example artifact

PASS | outputs/part-a/worked_example_por_0649.md contains raw input, initial facts, resolved dependencies, live trace, calculations, final conclusions, Maximum, and Union.

6. Part A - Detailed FV, CF, and CV Calculations

	Production Rule	FV, CF, and CV Calculations
R1	$\text{OR}(0.25, 1.0) = 1.0$	$\text{FV}=1.0; \text{CV}=1.0 \times 0.70 = 0.70$
R2	$\text{AND}(0.666667, 0.333333) = 0.333333$	$\text{FV}=0.333333; \text{CV}=0.333333 \times 0.90 = 0.300000$
R3	$\text{OR}(0.0, 0.70)=0.70; \text{AND}(0.30, 0.70)=0.30$	$\text{FV}=0.30; \text{CV}=0.30 \times 0.85 = 0.255$
R4	$\text{OR}(0.666667, 0.333333)=0.666667; \text{NOT}=0.333333; \text{AND}(0.25, 0.333333)=0.25$	$\text{FV}=0.25; \text{CV}=0.25 \times 0.60 = 0.15$
R5	$\text{NOT}(0.666667)=0.333333; \text{AND}(0.333333, 0.333333)=0.333333$	$\text{FV}=0.333333; \text{CV}=0.333333 \times 0.65 = 0.216667$
R6	$\text{AND}(0.0, 1.0)=0.0; \text{OR}(0.255, 0.0)=0.255$	$\text{FV}=0.255; \text{CV}=0.255 \times 0.90 = 0.2295$

6.3 Final Conclusions and Knowledge Base aggregation

Final Conclusion	Value
attendance_based_support_need	0.150000
emerging_performance_support_need	0.216667
compounded_academic_support_need	0.229500

Maximum = $\max(0.15, 0.216667, 0.2295) = 0.2295$

$U(0.15, 0.216667) = 0.15 + 0.216667 - (0.15 \times 0.216667) = 0.334167$

$U(0.334167, 0.2295) = 0.486975$

Final Union = 0.486975

The strongest individual conclusion is compounded academic support need. Union summarizes combined support across all Final Conclusions; it does not replace their distinct meanings.

7. Part B - Data Preparation

PART B - DATA PREPARATION

2 MARKS

VERIFIED

The selected source is the UCI Student Performance dataset, file student-por.csv. It contains Portuguese-language course records. The Mathematics and Portuguese files are not concatenated because 382 students are documented as shared and no true unique student ID is provided.

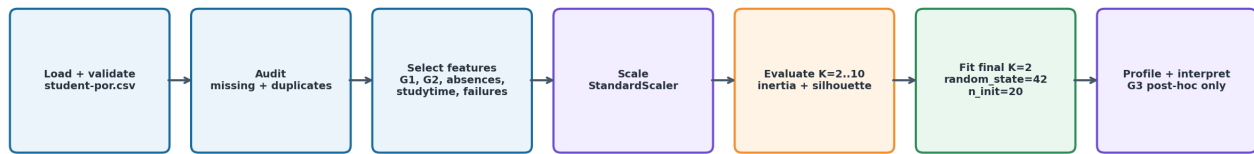
	Source	Notes
Raw shape	649 rows x 33 columns	Retain the approved Portuguese-course file
Delimiter	Semicolon	Explicit parser configuration
Missing cells / incomplete selected rows	0	No fabricated imputation
Empty strings	0	No replacement needed
Exact duplicates	0	No rows removed
Selected features	G1, G2, absences, studytime, failures	Five-domain-feature matrix
Encoding	Not required	Selected inputs are numeric / ordinal codes
Scaling	StandardScaler	Equalize distance contribution
G3	Excluded from training	Post-hoc interpretation only

Valid extreme values are retained rather than mechanically removed by IQR. The studytime field is treated as an ordered category code, not an exact hour count.

Data-preparation proof
PASS data/raw/uci_student_performance/student-por.csv is the approved source.
PASS Data audit and feature decisions are documented in docs/data/.
PASS data_preparation.py validates columns, delimiter, missingness, duplicates, selection, and scaling.
PASS G3 is absent from the training matrix and preserved only for post-hoc checks.

7. Part B - Pipeline and Preparation Pseudocode

Part B Reproducible K-Means Pipeline



Saved evidence: k_evaluation.csv | k_selection.png | cluster_profiles.csv | cluster_assignments.csv | summary.json

Figure 3. End-to-end Part B pipeline and saved evidence artifacts.

7.1 Preparation pseudocode

```

PREPARE_DATA(path):
data <- READ_CSV(path, delimiter=';')
REQUIRE expected UCI columns exist
duplicate_count <- COUNT_EXACT_DUPLICATES(data)
missing_count <- COUNT_MISSING(data[features + G3])
data <- DROP_EXACT_DUPLICATES_AND_INCOMPLETE_REQUIRED_ROWS(data)
X_raw <- data[G1, G2, absences, studytime, failures]
X_scaled <- StandardScaler.fit_transform(X_raw)
post_hoc_G3 <- data[G3] # never included in X_scaled
RETURN cleaned_data, X_raw, X_scaled, post_hoc_G3, audit_metadata
  
```

Why scaling matters: K-Means uses Euclidean distances. Without scaling, a feature with a larger numeric spread could dominate the geometry even when it is not more important to the domain.

Visual/output evidence

PASS | The executed notebook presents the same preparation stages and imports reusable tested modules.

PASS | summary.json records feature names, row counts, selected K, deterministic parameters, and data audit metadata.

8. Part B - Model Implementation

PART B - MODEL IMPLEMENTATION

2 MARKS

VERIFIED

The implementation uses scikit-learn KMeans. Candidate values K=2 through K=10 are evaluated on the standardized matrix. Every candidate and the final model use random_state=42 and n_init=20 for reproducibility.

8.1 K-Means pseudocode

```
EVALUATE_K(X_scaled):
results <- empty list
FOR k IN 2..10:
model <- KMeans(n_clusters=k, random_state=42, n_init=20)
labels <- model.fit_predict(X_scaled)
inertia <- model.inertia_
silhouette <- SILHOUETTE_SCORE(X_scaled, labels)
results.append(k, inertia, silhouette)
selected_k <- k with highest silhouette; use parsimony in close cases
RETURN results, selected_k

FIT_FINAL_MODEL(X_scaled, selected_k=2):
final <- KMeans(2, random_state=42, n_init=20).fit(X_scaled)
assignments <- final.labels_
profiles <- GROUP original-scale X_raw by assignments
names <- INTERPRET profiles only after fitting
SAVE assignments, profiles, metrics, chart, summary
```

8.2 Reproducible outputs

File Name	Description
outputs/ml/k_evaluation.csv	K=2..10 inertia and Silhouette evidence
outputs/ml/k_selection.png	Elbow and Silhouette visualization
outputs/ml/cluster_assignments.csv	One assignment for each of 649 retained rows
outputs/ml/cluster_profiles.csv	Original-scale cluster summaries
outputs/ml/summary.json	Parameters, selected K, audit, and reproducibility metadata
notebooks/part_b_kmeans_analysis.ipynb	Executable academic presentation layer

Model implementation proof

- PASS | Standard library implementation: scikit-learn KMeans.
- PASS | All 649 retained records receive exactly one cluster assignment.
- PASS | The final model is deterministic under fixed random_state and n_init.
- PASS | Reusable module code, runner, saved outputs, notebook, and tests are all present.

9. Part B - Evaluation and K Selection

PART B - EVALUATION

1 MARK

VERIFIED

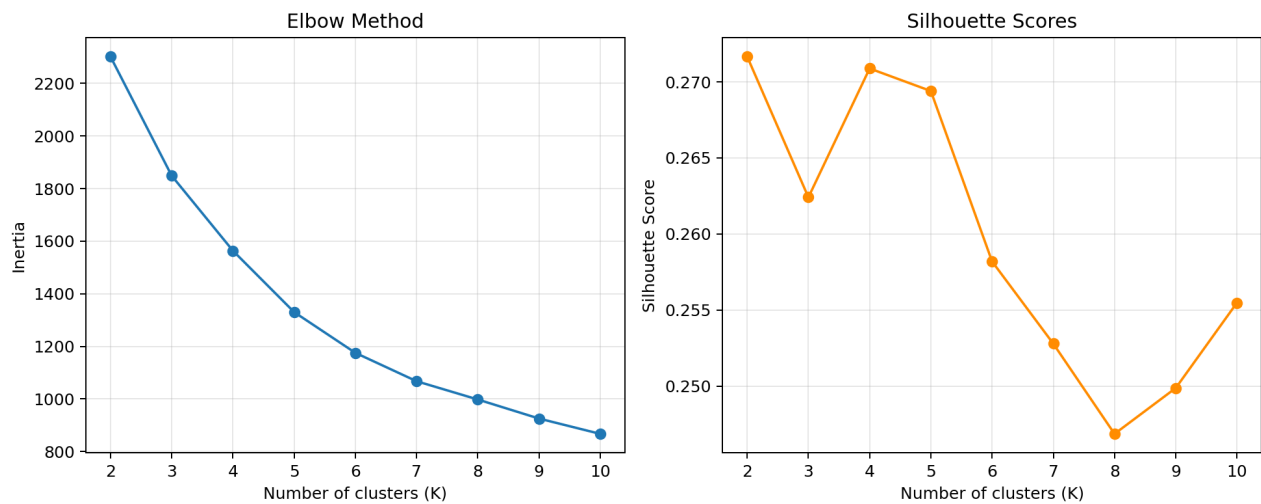


Figure 4. Actual pipeline output: inertia decreases gradually; K=2 has the highest measured Silhouette Score.

K	Inertia	Silhouette
2	2302.167	0.2717
3	1848.916	0.2624
4	1563.691	0.2709
5	1330.206	0.2694
6	1175.209	0.2582
7	1067.264	0.2528
8	997.714	0.2469
9	925.246	0.2499
10	866.953	0.2555

K=2 is selected because it has the highest observed Silhouette Score (0.2716646869) and is more parsimonious than the close K=4 candidate. The elbow is gradual, not uniquely sharp. The score indicates overlapping but meaningful profiles, not perfectly separated natural classes.

Evaluation proof

PASS | Both inertia and Silhouette Score are reported for every K from 2 to 10.

PASS | The actual saved chart visually supports the stated selection.

PASS | The interpretation explicitly avoids claiming that two objective truth classes exist.

9. Part B - Cluster Profiles and Interpretation

Cluster Name	Count	Mean Grade	Mean Studytime	Mean Absences	Mean Prior Failures	Mean Post-hoc G3	Mean Silhouette
Cluster 0: stronger academic profile	332	13.380	13.593	2.401	2.286	0.012	14.033
Cluster 1: higher support-need profile	317	9.325	9.451	4.978	1.558	0.442	9.678

Cluster names were assigned only after fitting and inspecting the original-scale profiles. Cluster 1 combines lower period grades, lower studytime code, more absences, and more prior failures. Its lower post-hoc G3 supports academic relevance, but G3 did not participate in scaling, K selection, or training.

9.1 Interpretation safeguards

- Clusters are descriptive groups, not ground-truth risk labels.
- Silhouette 0.2717 indicates overlap; this is not classification accuracy.
- G3 is a post-hoc check only, so the unsupervised feature matrix does not contain the final outcome.
- The dataset comes from two Portuguese schools and is not automatically generalizable to all institutions.

Cluster-output proof
PASS cluster_profiles.csv contains the observed original-scale means.
PASS cluster_assignments.csv contains all 649 row-level assignments.
PASS The notebook and KMEANS_RESULTS.md reproduce and explain the profiles.

10. Part A / Part B Integration

Integration is performed after both systems run independently. Part A conclusions are not K-Means features, and cluster IDs are not Expert System facts. Outputs are joined by source_row only for comparative analysis.

10.1 Integration pseudocode

```
RUN_INTEGRATION:
students <- LOAD same 649 cleaned records
clusters <- LOAD saved K=2 assignments
FOR each student:
part_a <- FUZZIFY_AND_RUN_FORWARD_INFERENCE(student)
record Final CVs, Maximum, Union
joined <- JOIN part_a outputs with clusters BY source_row
summary <- GROUP joined BY cluster:
mean Final CVs, contribution rates, mean Maximum, mean Union
correlate G3 with Maximum and Union only as post-hoc description
SAVE student comparison, cluster summary, chart, and metadata
```

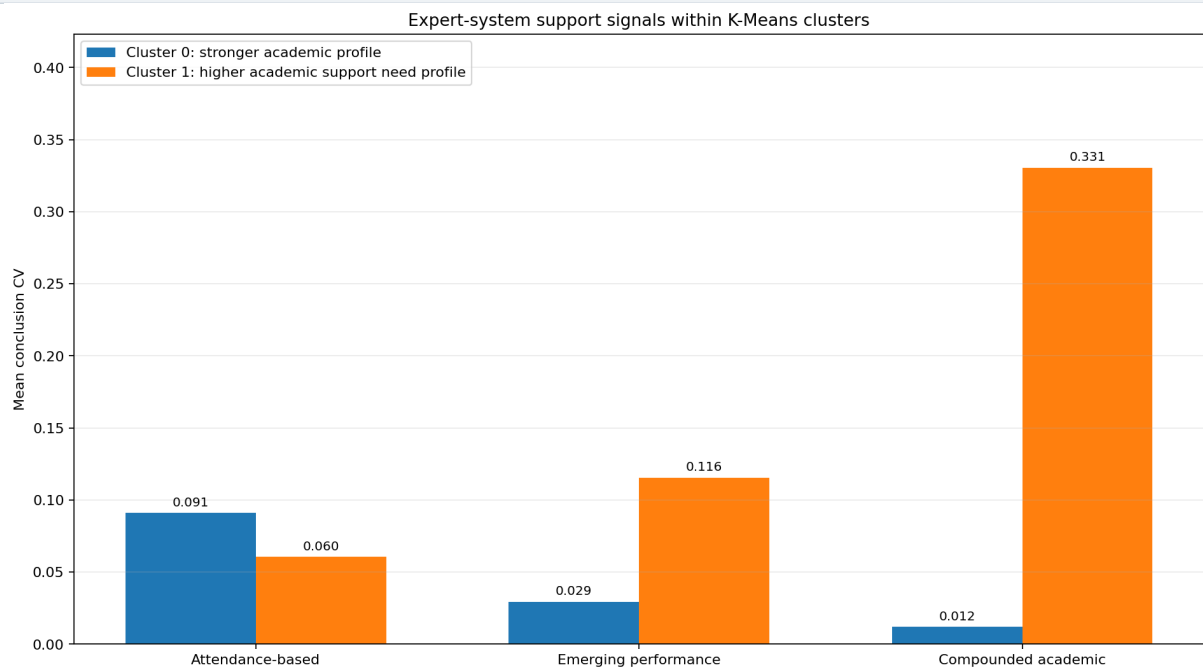


Figure 5. Actual integration output: mean Final-Conclusion CVs by independently discovered K-Means cluster.

10. Integration Results and Interpretation

Measure	Cluster 0: stronger profile	Cluster 1: higher support record
Students	332	317
Mean Maximum	0.118409	0.386468
Mean Union	0.128029	0.459942
Attendance contribution	31.33%	23.97%
Emerging-performance contribution	12.35%	42.90%
Compounded-support contribution	4.82%	86.12%
Post-hoc G3 mean	14.033	9.678

Cluster 1's mean Union is 3.59 times Cluster 0's. The largest separation is the chained compounded-support conclusion, showing broad directional agreement between explicit expert reasoning and unsupervised population patterns.

The attendance-based R4 signal is slightly stronger in Cluster 0. This is not hidden as a contradiction: R4 specifically models high absence when low-performance membership is limited, so it can identify an attendance-specific concern among otherwise stronger records.

Pearson correlations are -0.536862 between G3 and Maximum and -0.536041 between G3 and Union. These are descriptive associations, not causality and not classification accuracy.

Integration proof
PASS 649 students are joined by source_row with one K-Means assignment and independent Part A outputs.
PASS student_level_comparison.csv, cluster_expert_summary.csv, summary.json, and the chart are saved.
PASS Tests enforce the no-coupling boundary and validate all 649 joined rows.

11. Testing, Reproducibility, and Output Evidence

The merged development baseline passes 219 automated tests. Tests do not prove that every domain assumption is universally correct, but they provide strong evidence that the approved specification is implemented consistently and reproducibly.

Test area	Evidence covered
Raw input + fuzzification	Types, ranges, immutability, every boundary, dataset-wide five-fact conversion
Models + parser	O-A-V range, AST shapes, precedence, nesting, malformed syntax, forbidden IF/THEN
Knowledge Base	Required fields, CF, unknown facts, collisions, duplicate IDs/producers, Final Conclusions
Dependency graph	Edges, deterministic ordering, declaration ties, cycles, self-dependencies, structural status
Inference	Working Memory, min/max/NOT, FV/CF/CV, CV propagation, trace, Maximum/Union, POR-0649
Part B	Audit, scaling, K candidates, selected K, assignments, profiles, outputs, full pipeline
Integration	649-row join, summaries, no-coupling safeguards, generated artifacts

11.1 Reproducible commands

```
python -m pytest -q
python -m student_academic_advisor.expert_system.demo
python -m student_academic_advisor.ml_pipeline.run_pipeline
python -m student_academic_advisor.integration.run_analysis
python -m jupyter nbconvert --to notebook --execute --inplace notebooks/part_b_kmeans_analysis.ipynb
```

Execution evidence

PASS | Latest merged technical baseline: 219 tests passed.

PASS | Approved notebook checkpoint: 16 executed code cells, zero errors, IDs present on all cells.

PASS | Generated Part A, ML, and integration outputs are committed as report evidence.

12. Requirement Compliance Matrix

Requirement	Implementation	Evidence	Status
>=2 KR techniques	Production Rules + O-A-V facts	Schema/rule tables in Sections 2-3	PASS
Knowledge separate from inference	rules.json vs generic engine	Architecture Figure 1 + file paths	PASS
Nested + chained rules	R3/R4/R6; R1/R2->R3->R6	Inference Network Figure 2	PASS
Forward chaining + dependency handling	AST references, topological sort, cycles	Pseudocode + resolved order	PASS
Explain fired rules/order	Live trace model and report	POR-0649 trace table	PASS
Fuzzy operators, FV/CF/CV	min/max/1-x and CV propagation	Detailed rule calculations	PASS
Maximum and/or Union	Both implemented for Final CVs	POR-0649 aggregation	PASS
Required inference diagram	Official shapes and gates	Full-page Figure 2	PASS
Part B data preparation	Audit, five features, scaling, no leakage	Pipeline Figure 3 + audit table	PASS
K-Means implementation	scikit-learn, K=2..10, deterministic K=2	Saved model outputs	PASS
K justification + evaluation	Inertia + Silhouette	Actual K-selection Figure 4	PASS
Domain interpretation	Profiles + cautious G3 check + integration	Figures 4-5 + profile tables	PASS

Administrative and individual-assessment tasks are separate from technical implementation: the official Cover Sheet still requires final verified team entry/signatures, and every member must prepare for the individual viva marks.

12.1 Evidence policy

- A document statement alone is not treated as proof when executable or visual evidence exists.
- Visual requirements are evidenced by rendered diagrams/charts.
- Mathematical requirements are evidenced by the live POR-0649 calculation.
- Non-visual validation requirements are evidenced by exact paths, pseudocode, behavior, and tests.
- Optional Streamlit/deployment work is not counted as official rubric compliance.

13. Limitations, Ethics, and Safe Interpretation

Area	Limitation / Safeguard
Dataset scope	Two Portuguese secondary schools; results should not be generalized automatically to other institutions.
Membership functions	Documented author decisions based on the selected data context; not learned parameters or institutional policy.
Confidence factors	Author-assigned values; transparent and reviewable, but not statistically estimated probabilities.
K-Means geometry	Euclidean-distance clustering after scaling; overlapping profiles and no ground-truth labels.
Silhouette	0.2717 is the best tested relative score, not evidence of perfect separation.
G3	Excluded from premises/training; used only for post-hoc descriptive checks.
Integration	Shared concepts make association plausible; agreement does not prove causality or accuracy.
Advisor use	Support signal and explanation aid only; no automatic academic or disciplinary decision.

The project intentionally favors explainability and reproducibility over unsupported predictive claims. A reviewer can inspect every rule, premise, membership, calculation, cluster metric, and interpretation boundary.

Safe-claim checklist

- PASS | No classification accuracy is reported for unsupervised clusters.
- PASS | Cluster labels are descriptive and assigned after training.
- PASS | Correlation is not described as causation.
- PASS | The system does not diagnose or make final institutional decisions.

14. Conclusion

The Student Academic Advisor satisfies the required Expert System and Machine Learning components as one coherent project. Part A provides transparent individual reasoning through Production Rules, O-A-V fuzzy facts, validated condition ASTs, deterministic forward chaining, dependency resolution, cycle detection, fuzzy FV/CF/CV calculations, a live trace, and final aggregation. Part B provides audited data preparation, standardized features, reproducible K-Means implementation, K evaluation, visualization, and cautious profile interpretation.

The post-hoc integration shows broad agreement around compounded academic-support need while preserving a useful attendance-specific rule pattern. The two approaches therefore complement each other without being forced into one model or overstated as ground truth.

Final technical verdict
PASS Part A non-individual rubric evidence: complete.
PASS Part B non-individual rubric evidence: complete.
PASS Source, tests, notebook, generated outputs, worked example, and visual artifacts are reproducible.
PASS Remaining submission work is administrative release handling and individual viva preparation.

14.1 Optional enhancement boundary

A Streamlit interface may later provide an interactive presentation layer, but it is not an official requirement and must import the existing tested modules rather than duplicate rule, inference, clustering, or integration logic.

References

1. DSC 311, AI Skills - Course Project, official project description, Summer 2026.
2. DSC 311, official Group Project Cover Sheet, Summer 2026.
3. UCI Machine Learning Repository, Student Performance, DOI: <https://doi.org/10.24432/C5TG7T>.
4. Cortez, P. and Silva, A. Using Data Mining to Predict Secondary School Student Performance, 2008.
5. scikit-learn documentation: KMeans, StandardScaler, and Silhouette Score.

Repository evidence index

Source repository: <https://github.com/seif-ahmed-ib/student-academic-advisor>

Primary branch at the verified checkpoint: develop

Architecture: docs/project-management/ARCHITECTURE_DECISIONS.md (v3.4)

Compliance audit: docs/project-management/FINAL_COMPLIANCE_AUDIT.md

Master checklist: docs/project-management/MASTER_REQUIREMENTS_CHECKLIST.md (v2.0)

Final report source: docs/report/FINAL_REPORT.md

Appendix A - Deliverable and Output Inventory

Part A Knowledge Base	student_academic_advisor/expert_system/knowledge_base/rules.json
Part A implementation	raw_input.py, fuzzification.py, models.py, rule_parser.py, validation.py, dependency_graph.py, fuzzy_logic.py, working_memory.py, inference_engine.py, trace.py, aggregation.py
Part A worked output	outputs/part-a/worked_example_por_0649.md
Inference diagram	outputs/part-a/inference_network.png and inference_network.svg
Part B implementation	student_academic_advisor/ml_pipeline/data_preparation.py, clustering.py, reporting.py, run_pipeline.py
Part B notebook	notebooks/part_b_kmeans_analysis.ipynb
Part B outputs	k_evaluation.csv, k_selection.png, cluster_profiles.csv, cluster_assignments.csv, summary.json
Integration implementation	student_academic_advisor/integration/analysis.py, reporting.py, run_analysis.py
Integration outputs	student_level_comparison.csv, cluster_expert_summary.csv, mean_conclusion_cv_by_cluster.png, summary.json
Testing	tests/expert_system/, tests/ml_pipeline/, tests/integration/ - 219 passed
Documentation	docs/data/, docs/part-a/, docs/part-b/, docs/integration/, docs/project-management/, docs/report/

This inventory is designed for the live viva: each claim in the report can be traced to a concrete implementation, test, or generated artifact.

Appendix A.1 Report version note

This PDF is a professional submission edition assembled from the approved report source and committed output evidence. Numerical results are preserved from the tested pipeline. New architecture and pipeline figures are explanatory presentation diagrams and do not introduce new behavior.

Appendix B - Viva Quick Evidence Map

Question	Answer	Evidence
What are your two KR techniques?	Production Rules and O-A-V fuzzy facts; structurally distinct and separated from engine logic.	rules.json + models.py + Section 2
Why an AST?	Parse once, preserve precedence/nesting, reject malformed syntax, avoid eval().	rule_parser.py + R3/R4/R6
Why forward chaining?	Inputs are known facts; the engine derives support conclusions in dependency order.	dependency_graph.py + trace order
What is FV/CF/CV?	FV is condition activation, CF is author confidence, CV=FV x CF and is propagated.	POR-0649 Section 6
Why no 0.5 threshold?	Official design uses degrees; fired iff FV>0 and contributes iff CV>0.	inference_engine.py + trace
Why scale K-Means inputs?	Euclidean distance should not be dominated by a larger numeric scale.	data_preparation.py + Figure 3
Why K=2?	Highest tested Silhouette and parsimonious; elbow is gradual; K=4 is close but not better.	Figure 4 + k_evaluation.csv
Are clusters risk labels?	No. They are overlapping post-hoc descriptive profiles, not ground truth.	profiles + limitations
How do Parts A/B connect?	Same domain concepts, independent computation, post-hoc comparison by source_row.	integration Figure 5

Individual marks total 9/20. Every member should be able to explain the evidence above without reading the report verbatim.

End of report.

Presentation Layer - Not a Separate AI Model

The project now includes a Streamlit interface that presents the already-tested Part A, Part B, and integration modules. It does not replace the Expert System, retrain K-Means, or duplicate academic logic. The interface exists to make the live viva reproducible, readable, and visually traceable.

Boundary preserved

- Part A results are not used to train K-Means.
- K-Means cluster labels are not inserted as Expert-System facts.
- Integration remains a post-hoc descriptive comparison.

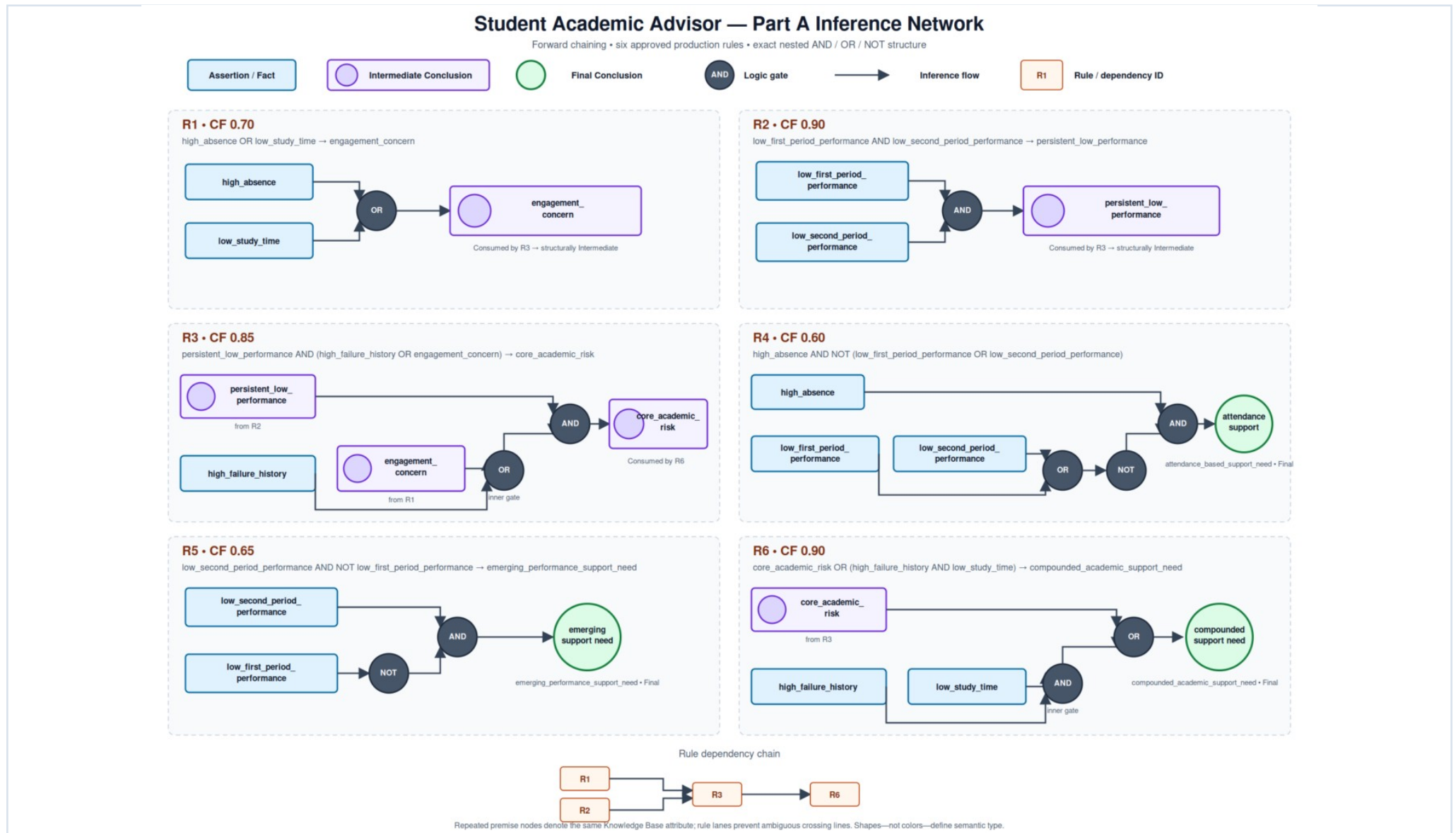
Interface tabs and purpose

Project Overview	High-level architecture, workflow, dataset size, selected K, and project boundary.
Individual Student Advisor	Interactive Part A inference with raw inputs, fuzzified facts, FV/CF/CV trace, final conclusions, Maximum, Union, and network diagram.
Part B - K-Means	Preparation evidence, StandardScaler, K=2..10 evaluation, inertia, Silhouette Score, selected K, and cluster profiles.
Part A / Part B Integration	Cluster-wise Expert-System signals, contribution rates, post-hoc G3 checks, and interpretation.
About / Sources / Limitations	Dataset scope, method boundaries, reproducibility commands, and limitations.

Run command

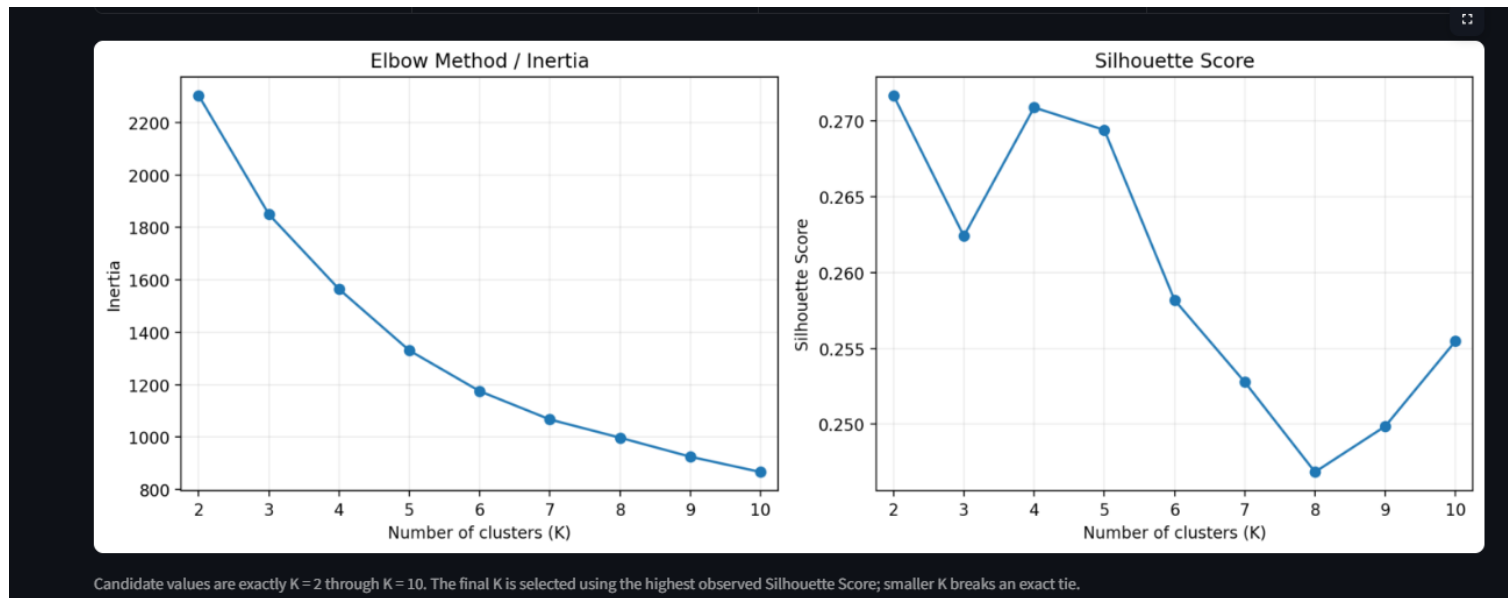
```
python -m streamlit run streamlit_app.py
```

Inference Network Diagram and rule dependency evidence

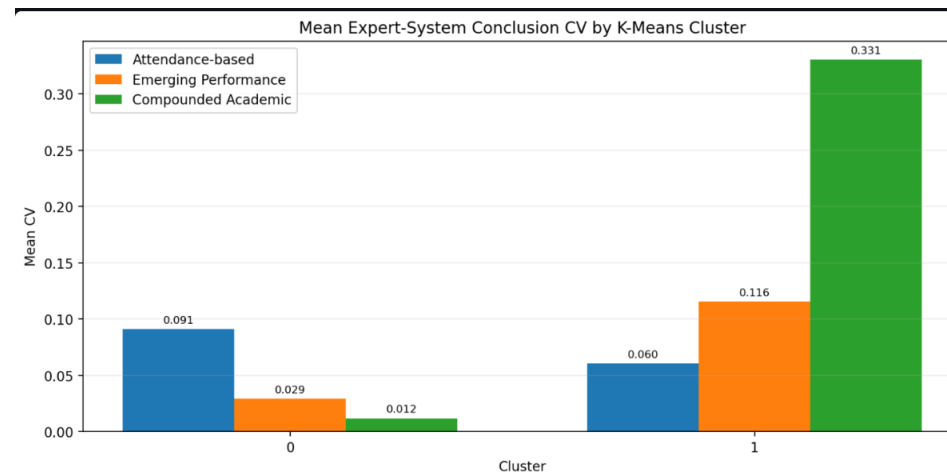


Evidence: exact six-rule network, Intermediate/Final distinction, AND/OR/NOT gates, and dependency chain R1,R2 -> R3 -> R6.

Part B: K-Means evaluation



Post-hoc integration: mean Final-Conclusion CV by K-Means cluster



Verified repository checkpoint

Branch	develop
Merge commit	1a19589 - Merge pull request #13 from feat/streamlit-presentation
Streamlit feature commit	58326d9 - feat: add Streamlit project presentation interface
Automated tests	219 passed
Working tree at verification	clean; develop aligned with origin/develop

Rubric-to-evidence summary

Rubric item	Implementation evidence	Visual / output evidence	Status
Part A - Knowledge representation	Production Rules + O-A-V; rules kept outside engine	Rule table + project structure	Verified
Part A - Inference engine	Forward chaining, dependencies, topological order, cycle detection, trace	Worked example + rule trace	Verified
Part A - Fuzzy logic	AND=min, OR=max, NOT=1-x, FV/CF/CV, Maximum/Union	Full POR-0649 calculations	Verified
Part A - Inference Network Diagram	Exact six-rule network	Diagram + Streamlit evidence	Verified
Part B - Data Preparation	649x33 source; five features; validation; StandardScaler; G3 excluded	Preparation table + UI evidence	Verified
Part B - Model Implementation	K-Means, random_state=42, n_init=20, K=2..10	Pipeline + outputs	Verified
Part B - Evaluation	Inertia + Silhouette; selected K=2; profile interpretation	K-selection plots + cluster profiles	Verified
Integration / coherence	Independent runs joined post-hoc by source_row	Cluster-vs-expert chart + correlations	Verified
Testing / reproducibility	Automated test suite and executable modules	219 passed + run commands	Verified
Presentation enhancement	Streamlit reuses tested modules	Five-tab presentation UI	Optional / Complete

Submission note

- The official Cover Sheet remains the administrative first page of the submission package.
- The core academic report remains unchanged in method and results; this appendix documents the completed presentation layer
- and the verified post-merge repository checkpoint.